

Read, Translate, Write: LLMs Translate with Fuzzy Semantic Hubs

Anonymous ACL submission

Abstract

[JB: main TODO is to kick off the BLEU ablation experiment for section 4, and tighten up the structure of the paper to accommodate] [JB: Rerun experiments for Aya] [JB: Describe the takeaways briefly, clarity is the most important. Don't aim for any length—summarizing the trends in a paragraph per table/figure is a rough estimate of what it might come to at the end] [JB: The methods could use a bit more detail of the methodology, explain to both a translation and mech interp audience, so explain both topics. Define the basic terms + cite the rest. Less detail needed for the prompting scheme.] [JB: Limitations can be done at the very end]

1 Introduction

Machine translation has long been an influential area of natural language processing. The attention mechanism was first proposed to improve machine translation performance (Bahdanau et al., 2015; Luong et al., 2015; Vaswani et al., 2017); one consequence of attention was the emergence of language models that could be efficiently trained on massive corpora (Devlin et al., 2019; Radford et al., 2018). So-called large language models (LLMs) have revolutionized many areas of natural language processing, but have been relatively slowly adopted in machine translation (MT).

A reason for the slow adoption of LLMs in MT has been a focus on low-resource languages, where LLMs face significant challenges (Hendy et al., 2023; Robinson et al., 2023). Because LLMs require large training corpora, they are difficult to train effectively on low-resource languages. However, they also hold significant promise: even when not trained on a given language, LLMs can be prompted to translate from or into a language they have seen very little of in their training data (Cahyawijaya et al., 2024)—e.g., by prompting

with a grammar book (Tanzer et al., 2024). Moreover, their representations of high-resource languages could enable cross-lingual transfer (Conneau et al., 2020) with little data in the target language.

Whether LLMs can deliver on this promise depends in part on the mechanisms underlying how they translate. Specifically, do LLMs rely on task- and language-agnostic mechanisms to translate, or are they more specialized to specific language pairs? If the former, this opens up a clear path toward improving MT performance, potentially without large parallel corpora (Artetxe et al., 2018). Much recent work provides mechanistic evidence supporting this hypothesis in the language modeling context (Wendler et al., 2024; Brinkmann et al., 2025; Wu et al., 2025). We therefore posit that LLMs might translate using the same mechanisms they use to language model: the “stages of inference” hypothesis (Lad et al., 2024) holds that LLMs devote the first half of their layers to reading an input and composing increasingly abstract concept representations. Then, the latter half of the model reads these concept representations, and uses them to decide which token should be predicted given prior context. This implies that translation would not necessarily require language pair-specific mechanisms: instead, it only requires that a model be capable of reading abstract features from the source language, and generating text for the target language conditioned on those features.

If this hypothesis is true, it has three main implications for MT. (1) Given language pair (S, T) , translation performance should be predictable from monolingual capabilities in S and T in isolation without cross-linguistic interaction terms. (2) There should exist task-agnostic representations that are causally relevant for predicting correct outputs in both machine translation *and* monolingual language modeling settings. Finally, (3) adding translation capabilities for a new language should

only require improvements to monolingual capabilities for that language—i.e., monolingual corpora should yield similar improvements in translation performance as parallel corpora, assuming the model can already effectively handle the other language in the language pair.

We investigate each of these three implications, and find positive evidence for each. While no single experiment is a smoking gun confirming the noisy channel hypothesis, each provides convergent evidence in support of it. This could provide a preliminary explanation for the value of exposure to (largely monolingual) documents in producing models more effective at translation.

[JB: Because of the updated experiments -> change noisy channel to something else that focuses on task agnosticism]

[JB: Jannik’s suggestion: add bolded contributions to this section]

2 Related Work

Massively multilingual feature representations. Our work is partially motivated by the observation that grammatical concept representations are highly multilingual in large language models, even across typologically distinct language pairs (Brinkmann et al., 2025). Similar evidence has been observed in (Wendler et al., 2024; Dumas et al., 2025).

Interpretability.

Low-resource machine translation.

3 A Motivating Observation: Monolingual Capabilities Explain Translation Performance

How much of a model’s ability to translate can be explained by its monolingual language modeling capabilities? We start by defining a **linear model** from monolingual competence to translation competence.

We use accuracies on MultiBLiMP as a proxy for monolingual competence. We take the macroaverage of scores across BLiMP categories for a given language.

For translation, we have language pair (S, T) , where S is the source and T is the target language. Translation competence t_{ST} is quantified as a BLEU score for (S, T) . For each language pair, we measure l_S and l_T , and learn coefficients β_S and β_T to maximize predictive accuracy in the

Model	R^2	MAE
Multi-BLiMP accuracy — linear	0.250	4.86
Multi-BLiMP accuracy — bilinear	0.251	4.84

Table 1: Table of some general results from 1

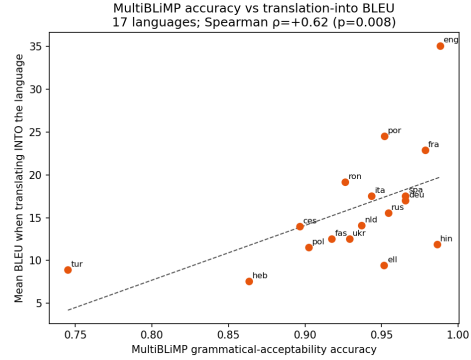


Figure 1: MultiBLiMP minimal pair accuracy vs. mean BLEU when translating into each language, over 17 Multi-BLiMP languages [JB: According to Claude, (ces, deu, ell, eng, fas, fra, heb, hin, ita, nld, pol, por, ron, rus, spa, tur, ukr). TODO: verify]. Spearman $\rho = 0.62$ ($p = 0.008$). [JB: Aaron notices that Multi-BLiMP might be saturated, and there’s lots of variability in BLEU even at high Multi-BLiMP scores]

following function:

$$\beta_S l_S + \beta_T l_T + \beta_0 = t_{ST} \quad (1)$$

We hypothesize that if a model is encoding an input into some universal feature space, then the predictive power of this model will be high. If a model relies on specialized translation mechanisms, the predictive power should be relatively low.

For further context, we also train a bilinear model with an interaction term:

$$\beta_S l_S + \beta_T l_T + \beta_{ST} l_{ST} = t_{ST} \quad (2)$$

If our hypothesis is correct, then the bilinear model should not have significantly greater predictive power than the linear model.

[AM: TODO: results]

[JB: I realized I have a table for this as well, let me copy it over]

[JB: TODO: add p values / Spearman correlation in 1]

[JB: TODO: add the 3 subplots with perplexity, linear model, bilinear model to the appendix. Gives visual interpretation of what the R scores mean.]



Figure 2: Llama BLEU matrix and its rank-1 SVD reconstruction. The missing cells along the diagonal are imputed by the mean of the target column. [JB: TODO: mask the diagonal out instead of imputation, I think this would overestimate the faithfulness] [JB: This is a little slippery, since the interaction effects may be marginalized, and can still be used in the rank1 approximation. Move to appendix]

4 Many Translation-Relevant Components Perform Monolingual Language Modeling

We turn our attention to show, mechanistically, that the model components implicated in translation are oftentimes those responsible for core language modeling in the monolingual setting.

4.1 Using GCM to Identify Translation-Relevant Components

We search for model components that are causal in promoting the valid translation over an invalid, but grammatically correct counterfactual.

Set $(p_{\text{orig}}, r_{\text{orig}}, r_{\text{cf}})$ where p_{orig} is the prompt with two candidate continuations r and define this preference metric:

$$M = \log \pi(r_{\text{cf}} | p_{\text{orig}}) - \log \pi(r_{\text{orig}} | p_{\text{orig}}), \quad (3)$$

where $\log \pi(r | p) = \sum_t \log p(r_t | p, r_{<t})$ is the teacher-forced log-probability of a continuation, summed over its tokens. M is thus the model’s log-odds preference for r_{cf} over r_{orig} under the same prompt. We then take a model component as a mediator; call its activation z . Then $M(z)$ becomes the above metric as a function of z when all other components are held constant. The change in M when moving from the counterfactual to the original [JB: Is this correct? I’m thinking about pos/neg here] is the indirect effect:

[JB: TODO: intuition about patching]

[JB: TODO: ensure that you really want negative indirect effect to mean model components that

influence the model to prefer the correct (grammatical) completion]

$$\text{IE}_{\text{patch}}(z) = M(z_{\text{orig}}) - M(z_{\text{cf}}). \quad (4)$$

[JB: Detail needed for GCM: this length is good, but make it actually well-understandable] [JB: The following a summary from Claude, do we keep something like this or refer to the paper?]

The GCM estimator. GCM (Sankaranarayanan et al., 2026) removes the per-component cost by linearizing M around z_{orig} . A first-order Taylor expansion gives $M(z_{\text{cf}}) \approx M(z_{\text{orig}}) + \nabla_z M|_{z_{\text{orig}}} \cdot (z_{\text{cf}} - z_{\text{orig}})$; substituting into Eq. 4,

$$\widehat{\text{IE}}(z) = \nabla_z M|_{z=z_{\text{orig}}} \cdot (z_{\text{orig}} - z_{\text{cf}}). \quad (5)$$

This recovers IE_{patch} to first order but is computable for *every* component at once: the gradients $\nabla_z M$ come from a single backward pass, and the deltas $(z_{\text{orig}} - z_{\text{cf}})$ from caching each component’s activation on the original and counterfactual runs. So one forward+backward pass yields an IE estimate for all 1024 heads and all 32,768 SAE features, versus one patched forward pass *each* for the exact version. [JB: End Claude]

The magnitude of the indirect effect is how much of the model’s preference flows through this component when all else is kept constant. Because we define $M = m_{\text{cf}} - m_{\text{orig}}$, components with a positive indirect effect push the model towards r_{cf} , and those with a negative indirect effect push the model towards r_{orig} . [JB: Maybe this is flipped?]

Now, we can set r_{orig} to be a gold translation of the original source, and r_{cf} a grammatically correct but invalid translation. We do this using the FLORES parallel corpus [JB: TODO: cite]. Given a source and target language, we construct a prompt as follows:

We patch at the last source token position only. For each of the 56 ordered cross-language pairs in {English, Spanish, German, French, Turkish, Arabic, Hindi, Hebrew}, we compute the IE for attention heads and SAE features.

However, this process could be simply isolating components that distinguish between the content of r_{orig} and r_{cf} . As control tasks, we modulate over two factors, where “Cross—Real” is the original experiment:

- Linguality:

178 “Same” refers to when $S = T$

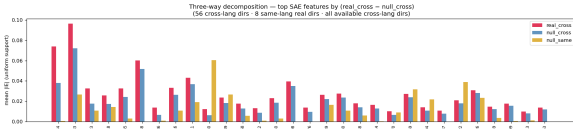


Figure 3: Comparison of GCM results and control tasks. These are the top SAE features as sorted by $\text{REAL_CROSS} - \text{NULL_CROSS}$. The bars represent the mean $|\widehat{\text{IE}}|$ of REAL_CROSS , NULL_CROSS , and NULL_SAME .

[JB: TODO: remove this, going with attn heads angle] [JB: TODO: idea that might not be worth it from Aaron—doing the same procedure but swapping the source instead of target, so that the control tasks are measuring off the same content] [JB: TODO: add interpretation—the blue bars being almost as high as the red bars indicate that there are many features which simply promote any sample in the target language, without necessarily promoting the *correct* translation]

– “Cross” is when translation is performed (i.e. the languages are distinct)

- Validity:

- “Real” indicates that we have chosen the valid translation as r_{orig}
- “Null” is when both r_{orig} and r_{cf} are random samples

[JB: TODO: decide on a format for real/null same/cross]

In particular, $\text{REAL_CROSS} - \text{NULL_CROSS}$ measures whether the component is in a translation-specific circuit or merely a multilingual one. NULL_SAME measures the pure content discrimination floor.

[JB: Here, I do think that the attention heads are not as discriminative as we’ve hoped, as $\text{REAL_CROSS} - \text{NULL_CROSS}$ is not that large of a difference for most heads. It does seem like SAE features do have a cleaner separation.]

[JB: TODO: Ablations of SAE features -> effect on BLEU Sort the SAE features by their real_cross . Progressively ablate more and more SAE features, measuring BLEU scores with more ablations]

4.2 Signedness of Indirect Effect of Heads Predicts Monolingual Grammatical Performance

[JB: TODO: in general, focus on attn heads, SAE stuff is on the back burner] [JB: TODO: cache some of the generations with and without ablations]

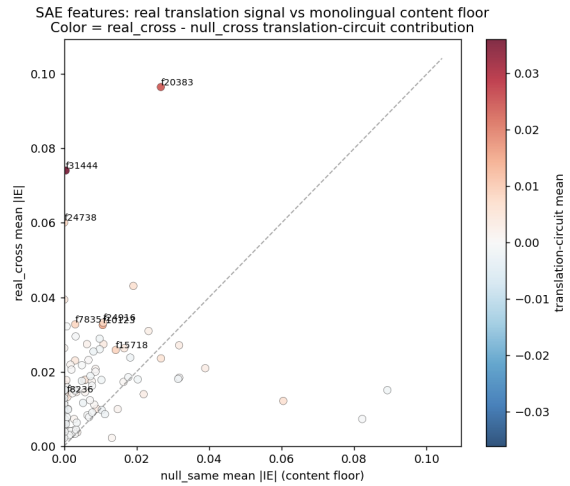


Figure 4: Here is the (REAL_CROSS) mean $|\widehat{\text{IE}}|$ plotted against the content-discrimination floor (NULL_SAME). Features well above the diagonal are active in translation contexts but not as much in monolingual settings. One such example is f31444, which had the largest absolute difference between REAL_CROSS and NULL_CROSS .

[JB: This could be largely subsumed by the other figures. Maybe it would be better to swap out null_same for null_cross and just have the plot be of that, with no color? OR swap for real_same , but real_same is kinda measuring copying a little bit instead of monolingual as we’d like it to]

We have found some model components responsible for differentiating between correct and incorrect translations. Is it possible that these same components are responsible for core language modeling?

[JB: On second thought, we should be trying some control task here where we ablate heads that were found in the null GCM experiments (i.e. still had high IE just not for translation)]

We use the Multi-BLiMP dataset, which contains minimal pairs of grammatical concepts, which measure the model’s ability to perform agreement tasks matching the subject to some target which must match in Person, Number, or Gender. We measure the model’s ability to distinguish between correct and incorrect completions. We take a random 400 (200 for Hebrew and Hindi) examples, resulting in a mix of tasks. An example:

[JB: TODO: This table 2 is from Claude, so will need to verify that this is in fact an actual example, plus clean up the width]

For each target language L in the GCM languages, we aggregate the GCM head indirect effects

Lang / phenomenon	Prefix	Ungram.
eng SV-P (Person 3→1)	Foris one thing, there	am
eng SV-# (Number Sg→Pl)	The verifies below site from the US government	verify
fra SV-P (Person 1→3)	Qui examinerai madame Schroedter, j'	examinera
fra SP-G (Gender M→F)	Combien de temps a	durée

Table 2: Representative MultiBLiMP minimal pairs. $\Delta > 0$ when the model prefers the grammatical continuation.

over the 7 directions where L is a distinct language. This allows us to select the top 20 heads by the mean $|\widehat{IE}|$, splitting the heads into POS and NEG categories based on their sign. Each Multi-BLiMP minimal pair is scored by the following:

$$\Delta = \log p(\text{correct completion} | \text{prefix}) - \log p(\text{incorrect completion} | \text{prefix}) \quad (6)$$

Now, we can compute the *difference in deltas* between the ablated and original models. That is, $\Delta_{\text{change}} = \Delta_{\text{ablation}} - \Delta_{\text{original}}$. The ablated model is produced by ablating the top 20 heads to their mean output at that position. A negative Δ_{change} means that the ablation weakened the model’s ability to distinguish between the correct and incorrect completions, the expected behavior.

[JB: TODO: write the results paragraph for this. As a placeholder, here’s something that Claude came up with:]

Result: GCM sign predicts the monolingual ablation effect. In all eight languages, ablating the NEG-signed sub-population *weakened* the grammaticality margin (Δ_{change} from -0.12 to -1.02), while ablating the POS-signed sub-population *strengthened* it ($+0.05$ to $+0.22$). A property measured purely in the translation task—the sign of a head’s GCM indirect effect—thus predicts the direction of its causal effect on a monolingual gram-

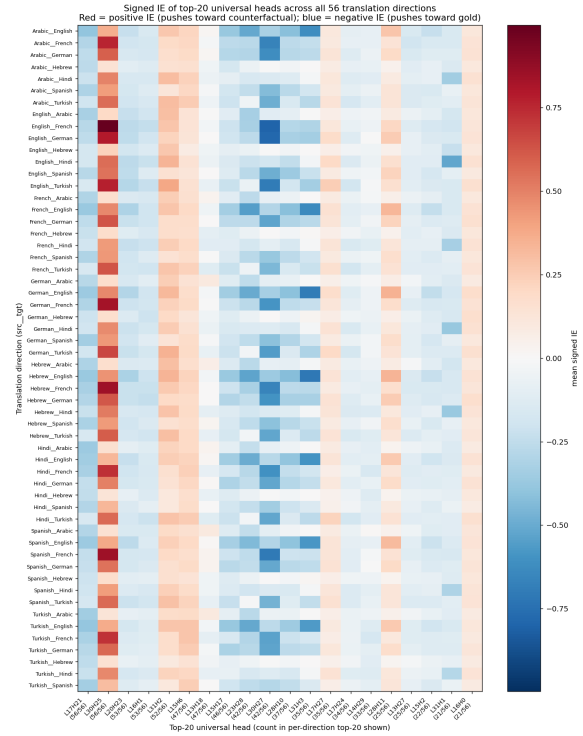


Figure 5: [JB: Two ideas here for an updated figure: we could get the top IE across all language pairs (in everything that we measured) or have some kind of multilinguality constraint as is present currently] Signed IE of the top 20 universal heads across all 56 cross-language directions. Red cells indicate positive IE (the head’s original state pushes toward the counterfactual response); blue cells indicate negative IE (pushes toward the gold response). The x-axis is sorted from left to right by the number of directions of which the head has a top 20 indirect effect.

[JB: The point of this is to show the cross-taskness and multilinguality. Include a condensed version of this (e.g. the top 14 rows)] [JB: Consider pushing this to the appendix]

maticity task. The NEG-signed heads (the majority, ~ 65 – 80% of the top-20) behave like shared language-modeling machinery that translation borrows; the smaller POS-signed population behaves like translation-specific routing, whose removal does not carry monolingual grammaticality information. By contrast, the mixed top-5 collective and the random controls are at the noise floor in 7/8 languages, because a set chosen by absolute IE alone mixes the two opposing populations. For the one language with size-matched controls so far (English—the only completed 53-condition v2 run), the matched-size POS/NEG random controls sit near zero (-0.028 , $+0.032$) while the sign-split groups are several times larger ($+0.075$, -0.160),

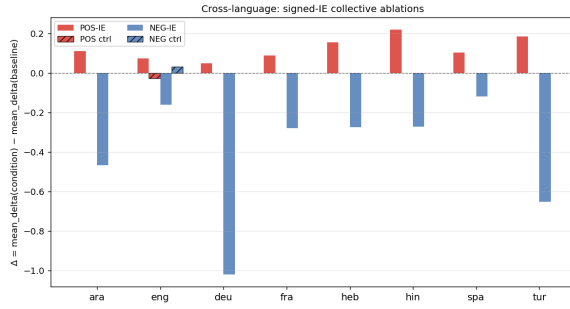


Figure 6: Per-language Δ_{change} on Multi-BLiMP from ablating the POS-signed vs. NEG-signed sub-population of the top-20 GCM heads. Negative means ablation weakened the grammaticality preference. Produced by visualize.py from analysis_summary.csv.

[JB: TODO: run the control ablations for non-English langs] [JB: Another worry: label how many heads are actually ablated for each pos, neg] [JB: Run this same experiment, but with BLEU and not Multi-BLiMP accuracy]

an early sign that the effect is not a group-size artifact; matched controls for the other seven languages are queued (job 5772145, n=400, A100). [JB: End Claude]

4.3 Grammatically-derived Features

[JB: This section is sparse because I’m not sure if we’d keep it. But we could give an account of how we derived the grammatical concepts SAE features (input and output) from UD, then show that there is some overlap between those SAE features and the GCM SAE features but Figure 7 has some positive evidence for the translation components being responsible for modeling the grammar]

5 Finetuning on monolingual data is competitive with finetuning on parallel corpora

If LLMs translate via language-general internal representations, translation quality should depend in part on the model’s monolingual competence in the source and target languages. Under this view, a translation error need not reflect a missing pair-specific mapping: it may instead reflect a failure to encode the source sentence into an adequate internal representation, or to decode the target sentence fluently from it. This suggests that improvements in monolingual language modeling should, at least in part, transfer to translation—even when using no parallel data.

We test this prediction by comparing monolin-

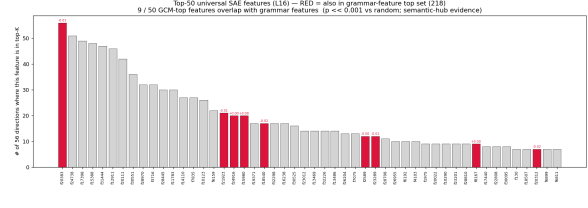


Figure 7: These are the 50 most direction-universal layer-16 SAE features, by number of the 56 directions in which each appears in the top-20. Red bars mark features that also belong to the grammatical concepts feature set identified in an earlier experiment. As opposed to the 0.33 expected by chance if the two sets were independent, 9/50 overlap.

[JB: TODO: verify the calculation needed to get 0.33] [JB: This is probably not very surprising/past results were muddy i.e. the ablations turned out to not have a large effect]

Model	Xho→Eng	Eng→Xho
Base	12.10	1.04
Best parallel FT	12.96	—
Best mono Xho FT	17.72	1.48

Table 3: BLEU for non-instruct Llama-3.1-8B LoRA runs. The best monolingual Xhosa run substantially outperforms the best parallel run found in our sweep on Xhosa→English.

[JB: TODO: check on the parallel FT for Eng → Xho]

[JB: Aaron has the idea of adding more language pairs here, and seeing if we could do better about eliciting the parallel data]

gual and parallel LoRA fine-tuning on the Llama-3.1-8B base model. We focus on Xhosa, where the base model has substantially weaker language modeling and translation performance than on other high-resource languages. The parallel runs train on 200K Xhosa–English sentence pairs; the monolingual runs train only on 200K samples of Xhosa text. For both settings, we run a small hyperparameter sweep over learning rate, LoRA rank, and number of training examples, and select the checkpoint with the highest Xhosa→English BLEU.

Translation quality. Table 3 reports BLEU for the base model, the strongest parallel run in our sweep, and the strongest monolingual run. The best parallel run yields a modest gain over the base model, from 12.10 to 12.96 Xhosa→English BLEU. The best monolingual run reaches 17.72 BLEU, substantially exceeding the strongest parallel adaptation we found despite using no translation supervision.

While we do not claim that monolingual finetuning dominates parallel fine-tuning in general,

Model	Xho PPL	Eng PPL	Fra PPL	Deu PPL
Base	66.70	21.86	11.25	12.50
Best parallel FT	71.04	23.42	11.20	13.96
Best mono Xho FT	26.23	40.63	34.72	34.05

Table 4: Perplexity for the same runs. Monolingual fine-tuning produces a much larger improvement on Xhosa, but degrades the non-Xhosa languages that parallel fine-tuning preserves.

we find that direct translation supervision was not necessary to obtain the improvements in translation performance. This is consistent with the hypothesis that gains in source-language competence transfer to translation via shared internal representations.

Perplexity. Table 4 clarifies the underlying trade-off. The monolingual run reduces Xhosa perplexity from 66.70 to 26.23, while the best parallel run leaves it essentially unchanged at 71.04. This supports the interpretation that the monolingual BLEU gain reflects improved Xhosa language modeling competence. The monolingual run, however, substantially degrades English, French, and German perplexity, whereas the parallel run preserves them.

Together, these results point to a specialization–preservation tradeoff: monolingual adaptation alone is sufficient to produce a large Xhosa→English gain, but it does so by specializing the model toward Xhosa at the expense of other languages. A practical recipe would likely combine monolingual adaptation with non-target-language replay to retain the translation gains without broader multilingual degradation.

6 Discussion and Conclusions

We have found that a linear model conditioned on proxies for monolingual competence is a strong predictor of translation performance, that many components are influential in both translation and monolingual tasks, and that continued pre-training on monolingual data often yields similar gains in performance to continued pre-training with multilingual data. Together, these act as convergent evidence in support of our main hypothesis.

We emphasize that mechanisms selective for the translation task are still likely to exist; indeed, [AM: TODO: Todd FV citation] find that there are components selective for word translation. It is likely that LLMs use a mixture of mechanisms [AM: TODO: cite mixture of mechanisms paper] to achieve their translation performance. Our goal was to explain why monolingual pretraining is helpful for transla-

tion at all, and not to explain the entirety of LLM translation performance.

Limitations

We’re only looking at 2 models of similar size

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A Appendix

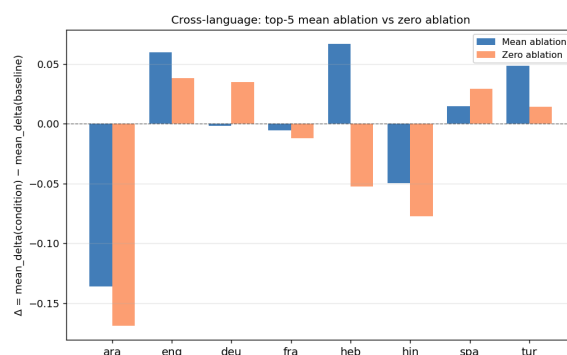


Figure 8: Mean vs. zero ablation

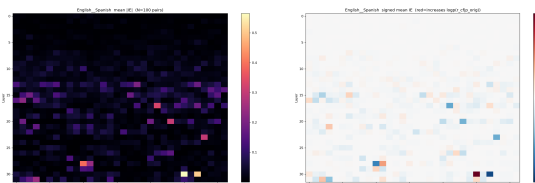


Figure 9: English→Spanish GCM

[JB: TODO: Inline you can say something about having a 2-shot prompt, but not much more elaboration is needed. Written by Claude:] The two shots are a fixed, held-out pair of FLORES sentences (rows 0–1, the same two for every direction); separators are single newlines and the final line ends in a trailing space, so the response begins as a bare target token:

English: On Monday, scientists from the Stanford University School of Medicine announced the invention of a new diagnostic tool that can sort cells by type: a tiny printable chip that can be manufactured using standard inkjet printers for possibly about one U.S. cent each.
 Spanish: El lunes, los científicos de la facultad de medicina de la Universidad de Stanford anunciaron el invento de una nueva herramienta de diagnóstico que puede catalogar las células según su tipo: un pequeñísimo chip que se puede imprimir y fabricar con impresoras de inyección de uso corriente, por un posible costo de, aproximadamente, un centavo de dólar por cada uno.
 English: Lead researchers say this may bring early detection of cancer, tuberculosis, HIV and malaria to patients in

	low-income countries, where	556
557	the survival rates for illnesses	
558	such as breast cancer can be	
559	half those of richer countries.	
560	Spanish: Los principales	
561	investigadores principales	
562	sostienen que esto puede	
563	permitir la detección precoz	
564	del cáncer, la tuberculosis, el	
565	VIH y la malaria en pacientes	
566	de países de bajos recursos,	
567	donde la tasa de supervivencia	
568	de enfermedades como el cáncer	
569	de mama puede ser la mitad de la	
570	de los países más avanzados.	
571	English: The JAS 39C Gripen	
572	crashed onto a runway at around	
573	9:30 am local time (0230 UTC)	
574	and exploded, closing the	
575	airport to commercial flights.	
576	Spanish:	
577	Under this single prompt we score two candi-	
578	date continuations: the gold target r_{orig} = “El	
579	JAS 39C Gripen impactó contra	
580	una pista cerca de las 9:30	
581	de la mañana hora local (0230	
	UTC) y explotó, lo que causó	582
	el cierre del aeropuerto para	583
	vuelos comerciales.” and a counterfac-	584
	tual r_{cf} = the gold translation of a different	585
	source, “Se identificó al piloto	586
	como Dilokrit Pattavee, líder de	587
	escuadrón.” (grammatical Spanish, but not a	588
	valid translation of the query). [JB: End Clauding]	589